



CS215 DISCRETE MATH

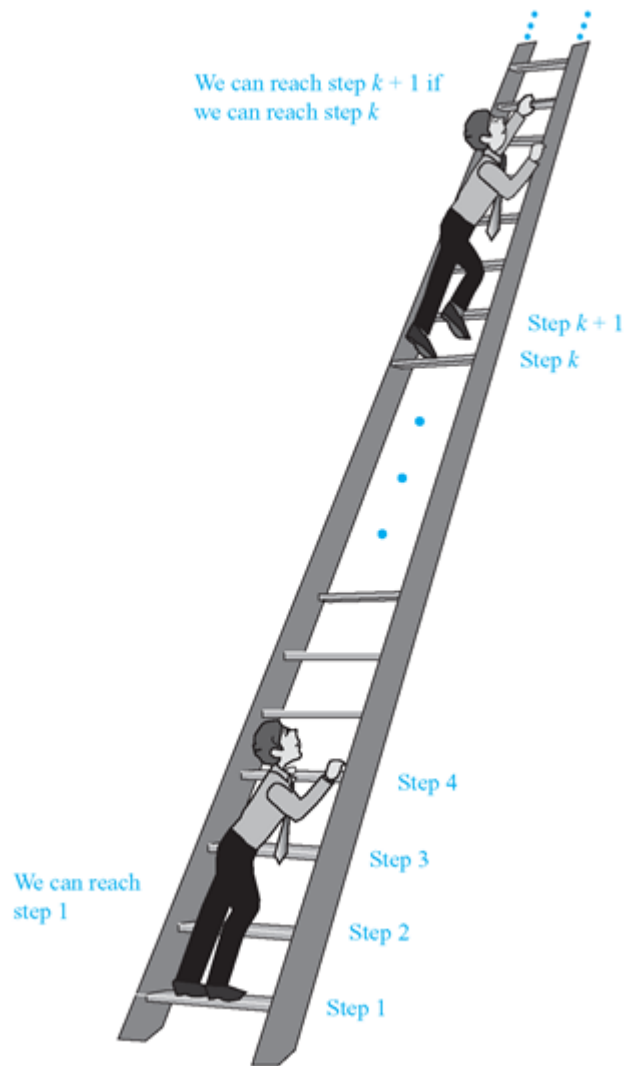
Dr. QI WANG

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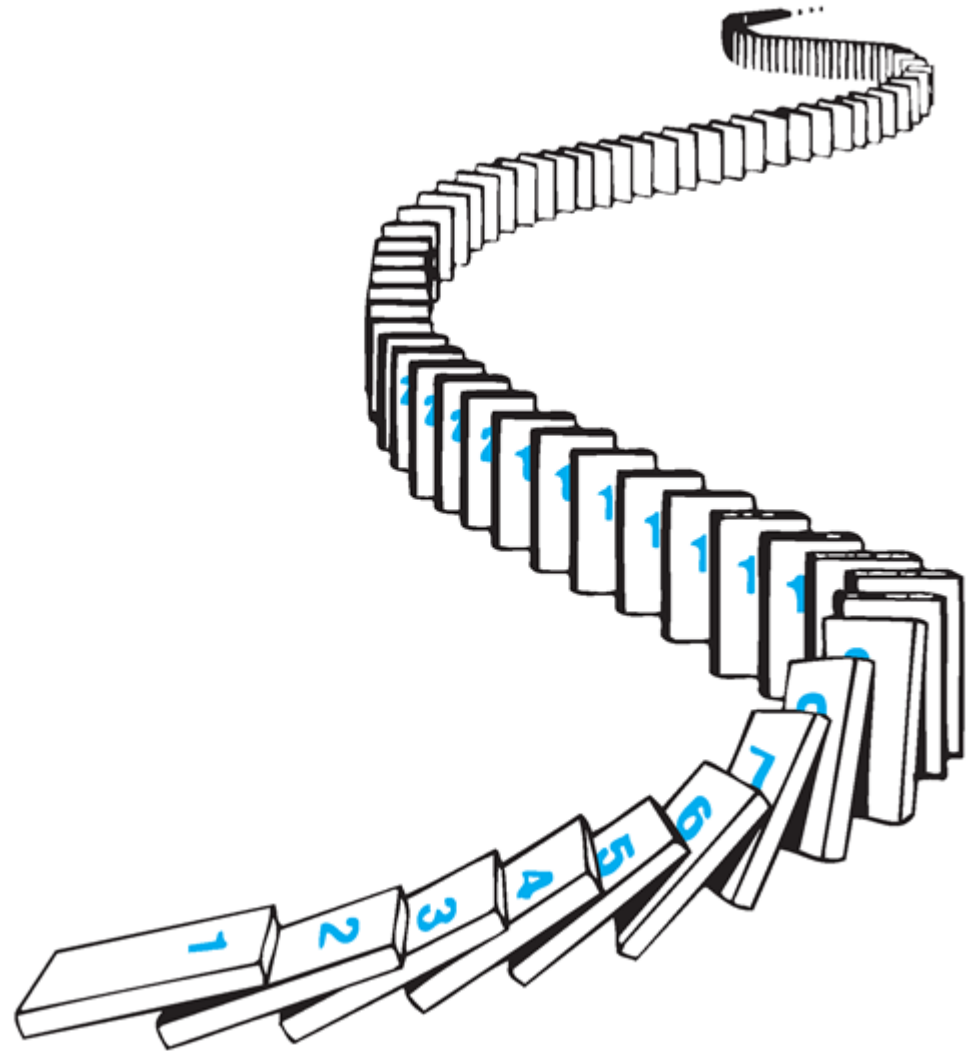
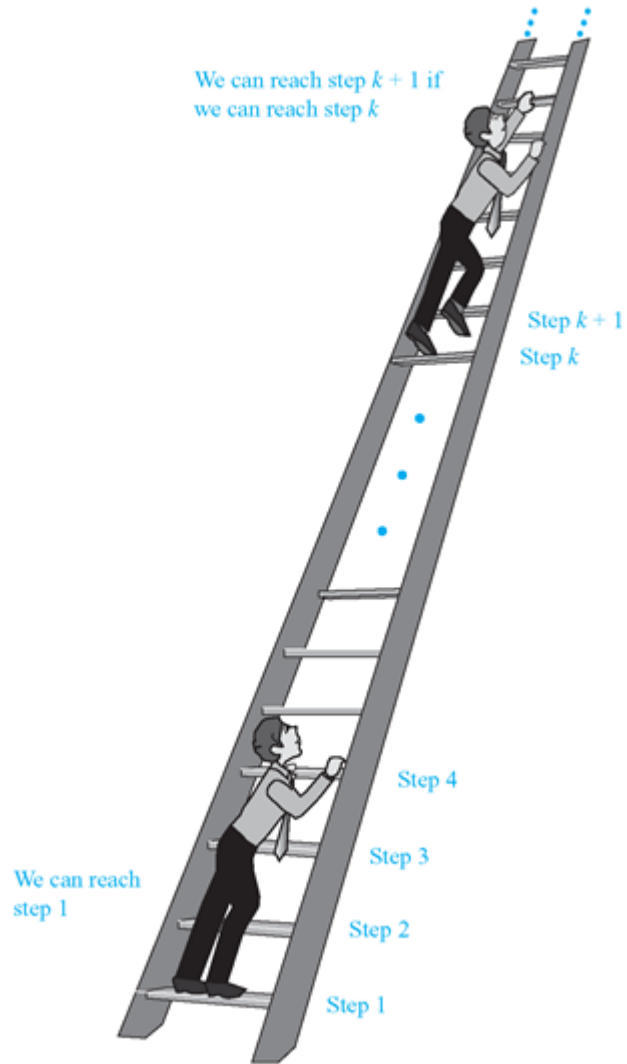
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- We conclude by distinguishing between the *weak principle* of mathematical induction and the *strong principle* of mathematical induction.



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- We conclude by distinguishing between the *weak principle* of mathematical induction and the *strong principle* of mathematical induction.

The *strong principle* can actually be derived from the *weak principle*.



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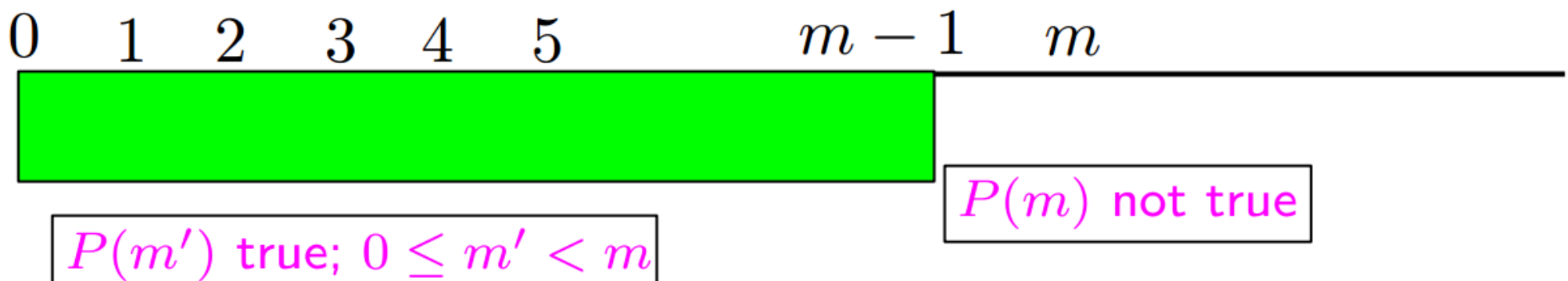


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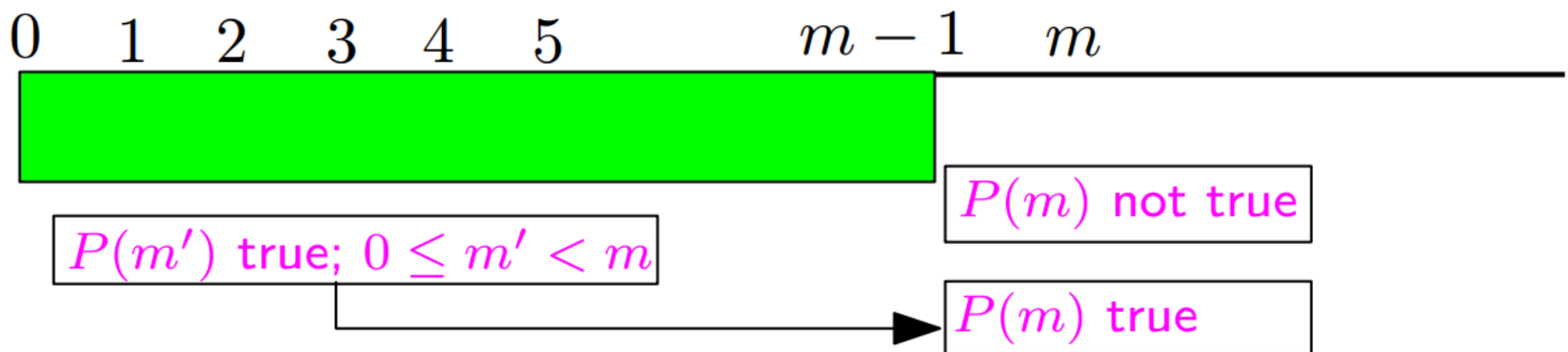


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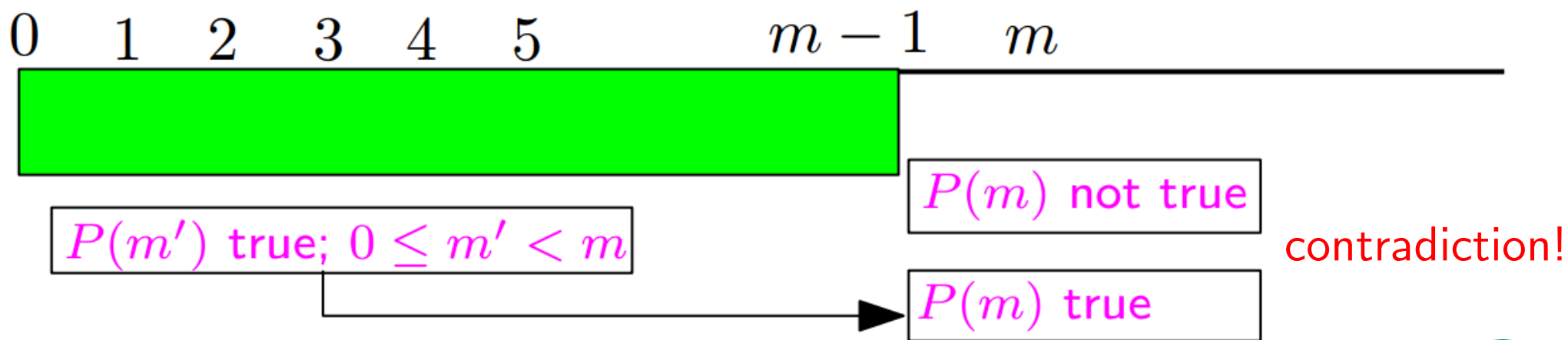


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- ◇ The smallest counterexample n is larger than 0



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◇ Therefore, $(*)$ holds for all positive integers n .



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The **key step** was proving that

$$P(n - 1) \rightarrow P(n)$$

where $P(n)$ is the statement

$$1 + 2 + \cdots + n = \frac{n(n + 1)}{2}$$

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Let $P(n) = 2^{n+1} \geq n^2 + 2$. We start by assuming that the statement

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is **false**.



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is **false**.

When a **for all** quantifier is false, **there must be some n for which it is false**. Let n be the **smallest nonnegative integer** for which $2^{n+1} \not\geq n^2 + 2$.

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This means that, for all $i \in \mathbb{N}$ with $i < n$,

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Since $2^{0+1} \geq 0^2 + 2$, we know that $n > 0$. Thus, **$n - 1$ is a nonnegative integer less than n .**



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Then setting $i = n - 1$ gives

$$2^{(n-1)+1} \geq (n-1)^2 + 2.$$

or

$$(*) \quad 2^n \geq n^2 - 2n + 1 + 2 = n^2 - 2n + 3$$



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Thus, we write

$$\begin{aligned} 2^{n+1} &\geq 2n^2 - 4n + 6 \\ &= (n^2 + 2) + (n^2 - 4n + 4) \\ &= n^2 + 2 + (n - 2)^2 \\ &\geq n^2 + 2. \end{aligned}$$



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10 - 5



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Since $P(n-1) \rightarrow P(n)$, we see that

$P(0)$ implies $P(1)$, $P(1)$ implies $P(2)$, ...



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Principle. (*the Weak Principle of Mathematical Induction*)

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(a) – *Basic Step* *Inductive Hypothesis*

(b) – *Inductive Step* *Inductive Conclusion*



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By mathematical induction, $\forall n > 0, 2^{n+1} \geq n^2 + 2$.



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$$\begin{aligned} 2^{n+1} &\geq 2(n-1)^2 + 6 \\ &= n^2 + 3 + n^2 - 4n + 4 + 1 \\ &= n^2 + 3 + (n-2)^2 + 1 \\ &> n^2 + 3 \end{aligned}$$

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By mathematical induction, $\forall n > 2, 2^{n+1} \geq n^2 + 3$.



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Base Step

(i) Note that for $n = 2, 2^{2+1} = 8 \geq 7 = 2^2 + 3 - P(2)$

(ii) Suppose that $n > 2$ and that $2^n \geq (n-1)^2 + 3 \quad (*)$

$$2^{n+1} \geq 2(n-1)^2 + 6 \quad \text{Inductive Hypothesis}$$

$$= n^2 + 3 + n^2 - 4n + 4 + 1$$

$$= n^2 + 3 + (n-2)^2 + 1$$

$$> n^2 + 3$$

Inductive Step

Hence, we've just prove that for $n > 2, P(n-1) \rightarrow P(n)$.

By mathematical induction, $\forall n > 2, 2^{n+1} \geq n^2 + 3$.

Inductive Conclusion



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- ◇ Iterating gives us a proof of $P(n)$ for all n



Strong Induction

- **Principle** (*The Strong Principle of Mathematical Induction*)
 - (a) If the statement $P(b)$ is true
 - (b) for all $n > b$, the statement $P(b) \wedge P(b+1) \wedge \cdots \wedge P(n-1) \rightarrow P(n)$ is true.then $P(n)$ is true for all integers $n \geq b$.



Example

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 - ◇ Then, if n is not a prime power, it is a product of two smaller numbers, each of which is, by the **inductive hypothesis**, a power of a prime or a product of powers of primes.
 - ◇ Thus, by the **strong principle of mathematical induction**, every positive integer is a power of a prime or a product of powers of primes.

Mathematical Induction

- In practice, we **do not** usually explicitly distinguish between the weak and strong forms.



Mathematical Induction

- In practice, we **do not** usually explicitly distinguish between the weak and strong forms.
- In reality, they are **equivalent** to each other in that **the weak form is a special case of the strong form, and the strong form can be derived from the weak form.**



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3. We conclude on the basis of the principle of **mathematical induction** that $P(n)$ is true for all $n \geq b$.

Recursion

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Recursion

- Recursive computer programs or algorithms often lead to *inductive analysis*.
- A classical example of *recursion* is the **Towers of Hanoi** Problem.



Towers of Hanoi



Towers of Hanoi



- 3 pegs; n disks of different sizes
- A *legal move* takes a disk from one peg and moves it onto another peg so that **it is not on top of a smaller disk**
- **Problem:** Find an (efficient) way to move all of the disks from one peg to another

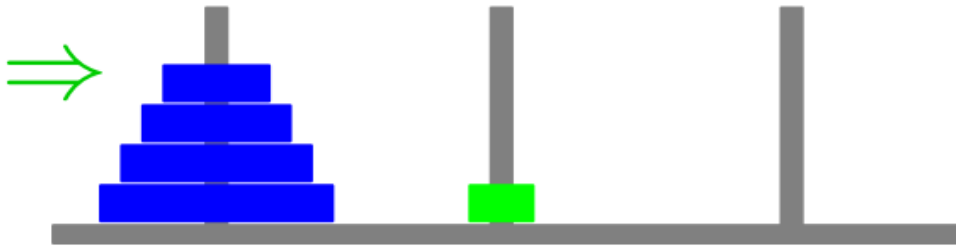
Towers of Hanoi



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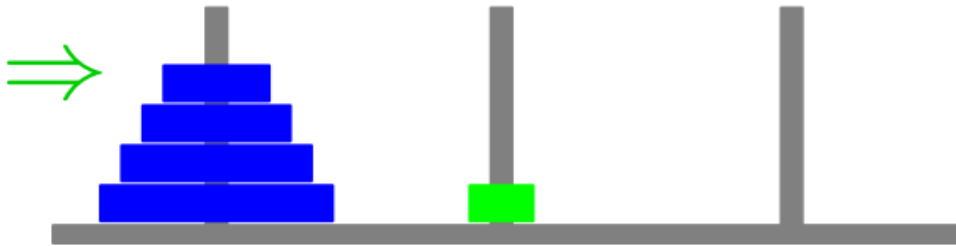
legal move



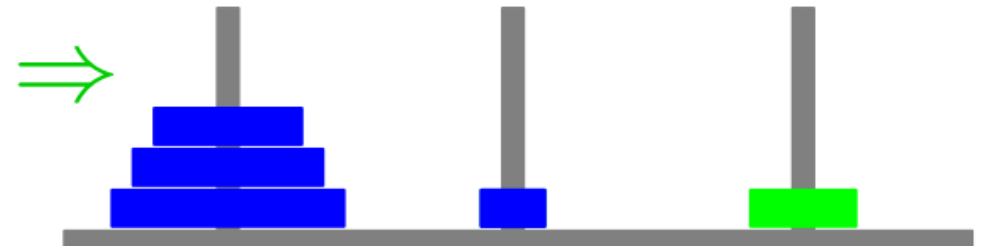
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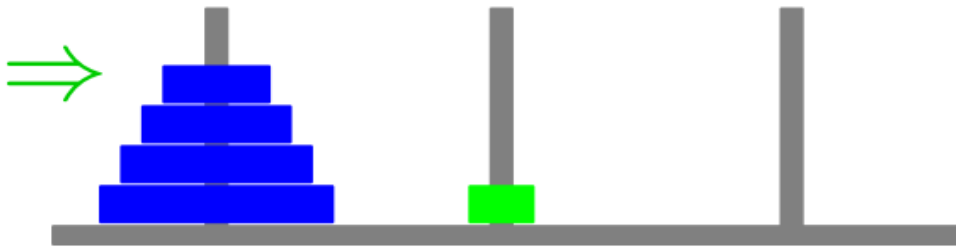
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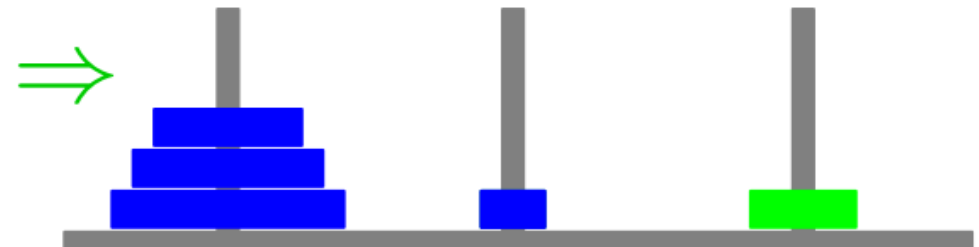
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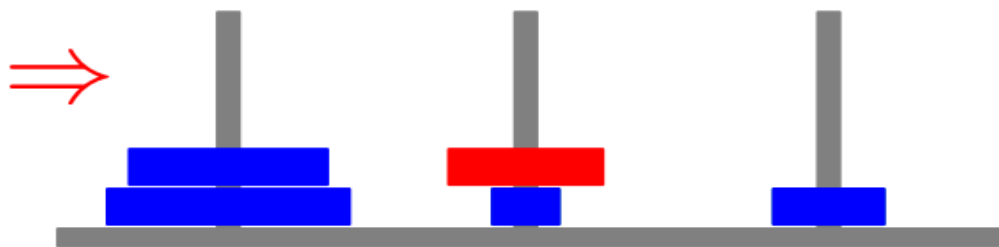
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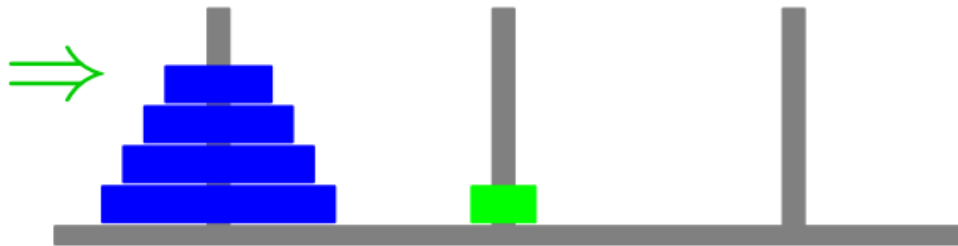
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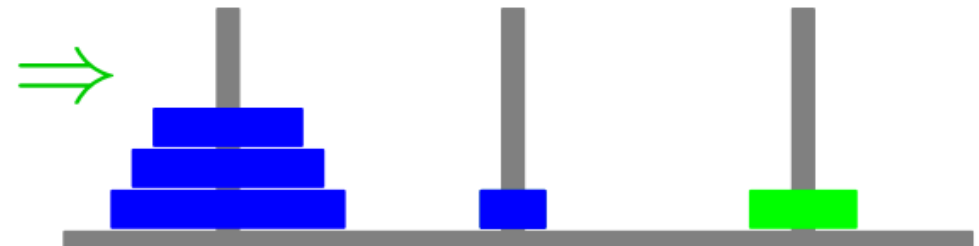
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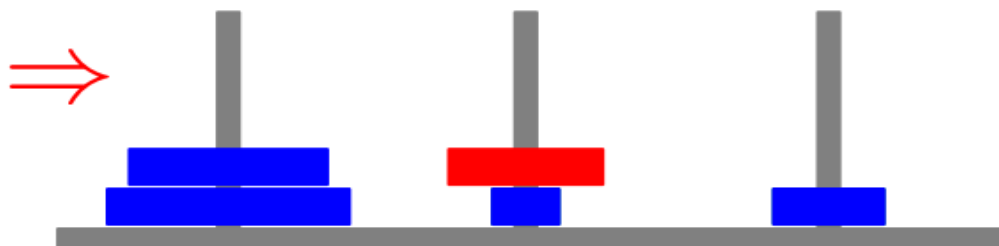
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Towers of Hanoi

- **Problem:** Start with n disks on leftmost peg



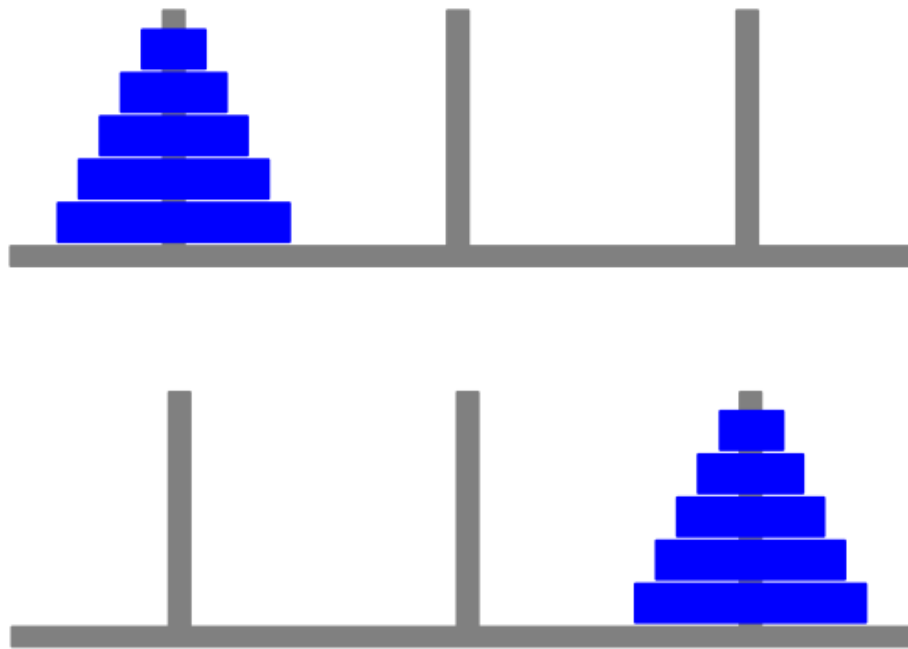
Towers of Hanoi

- **Problem:** Start with n disks on leftmost peg using only legal moves



Towers of Hanoi

- **Problem:** Start with n disks on leftmost peg using only legal moves
move all disks to rightmost peg.



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Given $i, j \in \{1, 2, 3\}$, let
 $\overline{\{i, j\}} = \{1, 2, 3\} - \{i\} - \{j\}$,
i.e., $\overline{\{1, 2\}} = \{3\}$, $\overline{\{1, 3\}} = \{2\}$,
 $\overline{\{2, 3\}} = \{1\}$.



Towers of Hanoi

- General solution



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Recursion Base:

If $n = 1$, moving one disk from i to j is easy. Just move it.



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Towers of Hanoi



To move $n > 1$ disks from i to j

Towers of Hanoi



To move $n > 1$ disks from i to j

1)



move top $n - 1$ disks from i to $\overline{\{i, j\}}$

Towers of Hanoi



1)



2)



To move $n > 1$ disks from i to j

move top $n - 1$ disks from i to $\overline{\{i, j\}}$

move **largest** disk from i to j

Towers of Hanoi



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3)



To move $n > 1$ disks from i to j

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move largest disk from i to j

move top $n - 1$ disks from $\overline{\{i, j\}}$ to j

Towers of Hanoi

```
3 public class Hanoi
4 {
5
6     public void move(int n, char a, char b, char c)
7     {
8         if (n == 1)
9             System.out.println("plate " + n + " from " + a + " to " + c);
10        else
11        {
12            move(n-1,a,c,b);
13            System.out.println("plate " + n + " from " + a + " to " + c);
14            move(n-1,b,a,c);
15        }
16    }
17 }
18
```



Towers of Hanoi

To move n disks from i to j

i) move top $n - 1$ disks from i to $\overline{\{i, j\}}$

ii) move largest disk from i to j

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Towers of Hanoi

- To prove **correctness** of solution, we are implicitly using **induction**

To move n disks from i to j

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- $p(n)$ is statement that algorithm is correct for n

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- $p(n - 1) \rightarrow p(n)$ is **recursion** statement that

if our algorithm works for $n - 1$ disks, then we can build a correct solution for n disks

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Towers of Hanoi

■ Running time

$M(n)$ is number of disk moves needed for n disks

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$$M(1) = 1$$

$$\text{if } n > 1, \text{ then } M(n) = 2M(n - 1) + 1$$



Towers of Hanoi

- We saw that $M(1) = 1$ and that
- $M(n) = 2M(n - 1) + 1$ for $n > 1$



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Later, we'll also see how to solve *without guessing*



Towers of Hanoi

- Formally, given

$$M(n) = \begin{cases} 1 & \text{if } n = 1 \\ 2M(n-1) + 1 & \text{otherwise} \end{cases}$$

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Then $M(n) = 2M(n-1) + 1 = 2(2^{n-1} - 1) + 1 = 2^n - 1$



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$$M(n) = \begin{cases} 1 & \text{if } n = 1 \\ 2M(n-1) + 1 & \text{otherwise} \end{cases}$$

- The second time was to **derive** the **closed form solution** $M(n) = 2^n - 1$ of the recurrence.



Recurrences

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$$M(n) = \begin{cases} 1 & \text{if } n = 1 \\ 2M(n - 1) + 1 & \text{otherwise} \end{cases} \quad \text{Towers of Hanoi}$$

$$F(n) = \begin{cases} 1 & \text{if } n = 0, 1 \\ F(n - 1) + F(n - 2) & \text{otherwise} \end{cases} \quad \text{Fibonacci Sequence}$$



Recurrences

- **Example 2:** Let $S(n)$ be the number of subsets of a set of size n . What is the formula for $S(n)$?

The empty set, of size $n = 0$ has only one subset (itself), so $S(0) = 1$.

It is not difficult to see that

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We “guess” that $S(n) = 2^n$. But, in order to prove formula, we’ll need to think recursively.



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This suggests that the recurrence for the number of subsets of an n -element set $\{1, 2, \dots, n\}$ is

$$S(n) = \begin{cases} 1 & \text{if } n = 0 \\ 2S(n-1) & \text{if } n \geq 1 \end{cases}$$



Recurrences

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Each subset S containing n can be constructed in a unique fashion by adding n to the subset $S - \{n\}$ not containing n .

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Proof by induction is easy.



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Can we generalize this to find a **closed-form solution**?



Iterating a Recurrence

- Note that $T(n) = rT(n-1) + a$ implies that
 $\forall i < n, T(n-i) = rT((n-i)-1) + a$



Iterating a Recurrence

- Note that $T(n) = rT(n-1) + a$ implies that

$$\forall i < n, T(n-i) = rT((n-i)-1) + a$$

Then, we have

$$\begin{aligned} T(n) &= rT(n-1) + a \\ &= r(rT(n-2) + a) + a \\ &= r^2 T(n-2) + ra + a \\ &= r^2(rT(n-3) + a) + ra + a \\ &= r^3 T(n-3) + r^2 a + ra + a \\ &= r^3(rT(n-4) + a) + r^2 a + ra + a \\ &= r^4 T(n-4) + r^3 a + r^2 a + ra + a. \end{aligned}$$

Iterating a Recurrence

- Note that $T(n) = rT(n-1) + a$ implies that

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Guess $T(n) = r^n T(0) + a \sum_{i=0}^{n-1} r^i$



Iterating a Recurrence

- The method we used to guess the solution is called *iterating the recurrence*, because we repeatedly (iteratively) use the recurrence.



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$$T(0) = b$$

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This would lead to the same guess

$$T(n) = r^n b + a \sum_{i=0}^{n-1} r^i.$$



Formula of Recurrences

- **Theorem** If $T(n) = rT(n-1) + a$, $T(0) = b$, and $r \neq 1$, then

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Proof by induction

The base case:

$$T(0) = r^0 b + a \frac{1 - r^0}{1 - r} = b.$$

So the formula is true when $n = 0$.

Now assume that $n > 0$ and

$$T(n-1) = r^{n-1} b + a \frac{1 - r^{n-1}}{1 - r}.$$



Formula of Recurrences

■ Proof by induction

$$\begin{aligned}T(n) &= rT(n-1) + a \\&= r \left(r^{n-1}b + a \frac{1-r^{n-1}}{1-r} \right) + a \\&= r^n b + \frac{ar - ar^n}{1-r} + a \\&= r^n b + \frac{ar - ar^n + a - ar}{1-r} \\&= r^n b + a \frac{1-r^n}{1-r}.\end{aligned}$$

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Plugging $r = 3$, $a = 2$, $b = 5$ in the formula, gives

$$T(n) = 3^n \cdot 5 + 2 \frac{1 - 3^n}{1 - 3} = 3^n \cdot 6 - 1$$



First-Order Linear Recurrences

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Something like $T(n) = (T(n-1))^2 + 3$ would be a **non-linear** first-order recurrence relation.



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First-Order Linear Recurrences

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When $f(n)$ is a constant, say r , the general solution is almost as easy as we derived before. Iterating the recurrence gives

$$\begin{aligned}T(n) &= rT(n-1) + g(n) \\&= r(rT(n-2) + g(n-1)) + g(n) \\&= r^2T(n-2) + rg(n-1) + g(n) \\&= r^3T(n-3) + r^2g(n-2) + rg(n-1) + g(n) \\&\vdots \\&= r^nT(0) + \sum_{i=0}^{n-1} r^i g(n-i)\end{aligned}$$



First-Order Linear Recurrences

- **Theorem** For any positive constants a and r , and any function g defined on nonnegative integers, the solution to the first-order linear recurrence

$$T(n) = \begin{cases} rT(n-1) + g(n) & \text{if } n > 0 \\ a & \text{if } n = 0 \end{cases}$$

is

$$T(n) = r^n a + \sum_{i=1}^n r^{n-i} g(i).$$



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$$\begin{aligned}T(n) &= 6 \cdot 4^n + \sum_{i=1}^n 4^{n-i} \cdot 2^i \\&= 6 \cdot 4^n + 4^n \sum_{i=1}^n 4^{-i} \cdot 2^i \\&= 6 \cdot 4^n + 4^n \sum_{i=1}^n \left(\frac{1}{2}\right)^i \\&= 6 \cdot 4^n + \left(1 - \frac{1}{2^n}\right) \cdot 4^n \\&= 7 \cdot 4^n - 2^n.\end{aligned}$$

Examples

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Theorem. For any real number $x \neq 1$,

$$\sum_{i=1}^n ix^i = \frac{nx^{n+2} - (n+1)x^{n+1} + x}{(1-x)^2}.$$

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$$\begin{aligned}T(n) &= 10 \cdot 3^n + \sum_{i=1}^n 3^{n-i} \cdot i \\&= 10 \cdot 3^n + 3^n \sum_{i=1}^n i \cdot 3^{-i} \\&= 10 \cdot 3^n + 3^n \left(-\frac{3}{2}(n+1)3^{-(n+1)} - \frac{3}{4}3^{-(n+1)} + \frac{3}{4} \right) \\&= \frac{43}{4}3^n - \frac{n+1}{2} - \frac{1}{4}.\end{aligned}$$

Growth Rates of Solutions to Recurrences

- Divide and conquer algorithms
- Iterating recurrences
- Three different behaviors

Divide and conquer algorithms

- We just analyzed recurrences of the form

$$T(n) = \begin{cases} b & \text{if } n = 0 \\ r \cdot T(n-1) + a & \text{if } n > 0 \end{cases}$$



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$$T(n) = r^n b + a \frac{1 - r^n}{1 - r}$$

We will now look at recurrences of the form

$$T(n) = \begin{cases} \text{something given} & \text{if } n \leq n_0 \\ r \cdot T(n/m) + a & \text{if } n > n_0 \end{cases}$$



Binary Search

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- ◇ Is x greater than k ?
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Our strategy will be to always ask greater than questions, at each step halving our search range, until the range only contains one number, when we ask a final equal to question.



Binary Search Example

1

32

48

64

52 - 1



Binary Search Example

1

32

48

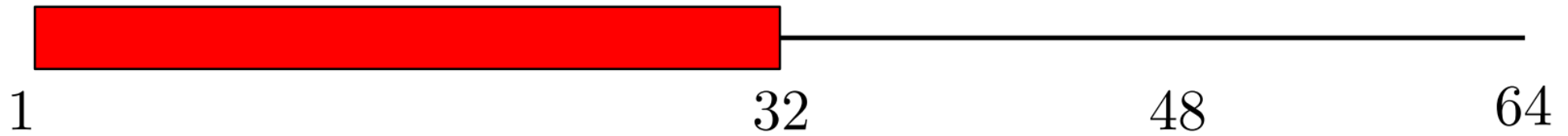
64

Is $x > 32$?

52 - 2



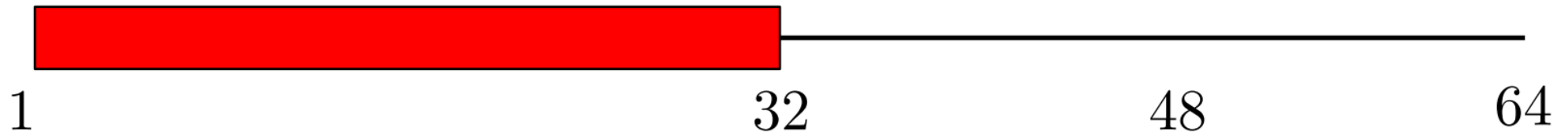
Binary Search Example



Is $x > 32$?

Answer: Yes

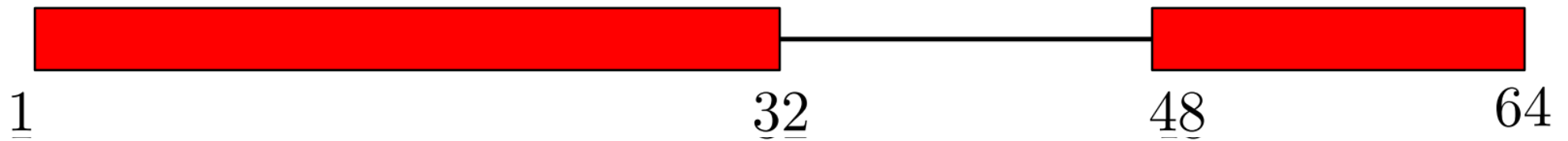
Binary Search Example



Is $x > 32$? Answer: Yes

Is $x > 48$?

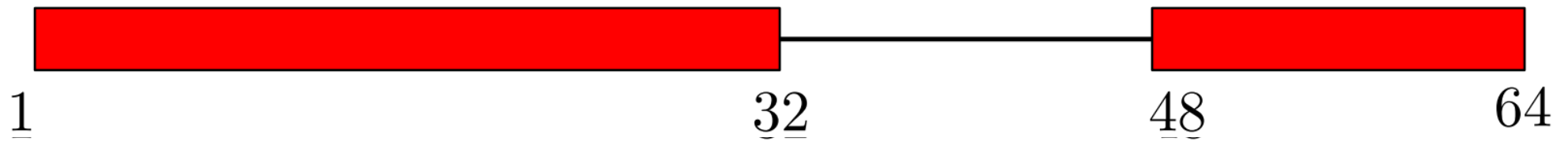
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Binary Search Example

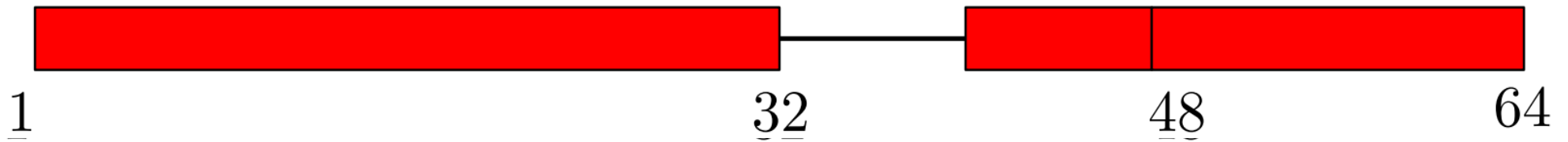


Is $x > 32$? Answer: Yes

Is $x > 48$? Answer: No

Is $x > 40$?

Binary Search Example

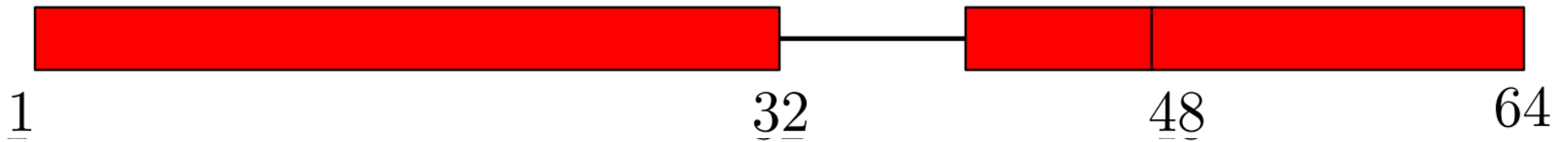


Is $x > 32$? Answer: Yes

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Binary Search Example



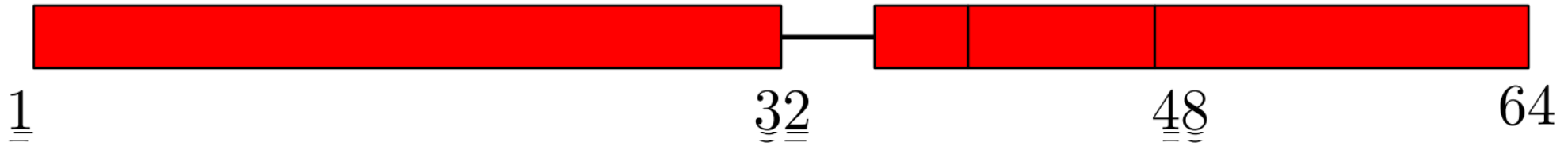
Is $x > 32$? Answer: Yes

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Is $x > 36$?

Binary Search Example



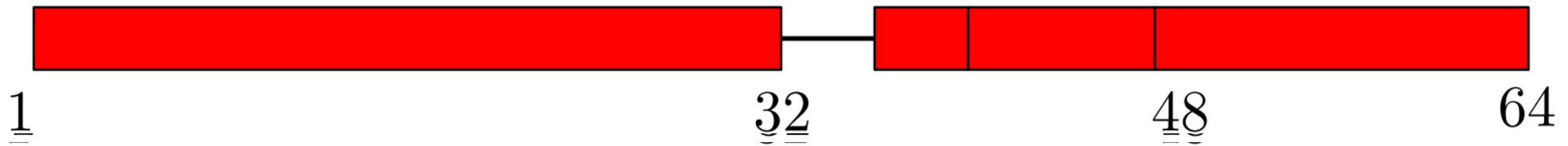
Is $x > 32$? Answer: Yes

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Is $x > 40$? Answer: No

Is $x > 36$? Answer: No

Binary Search Example



Is $x > 32$? Answer: Yes

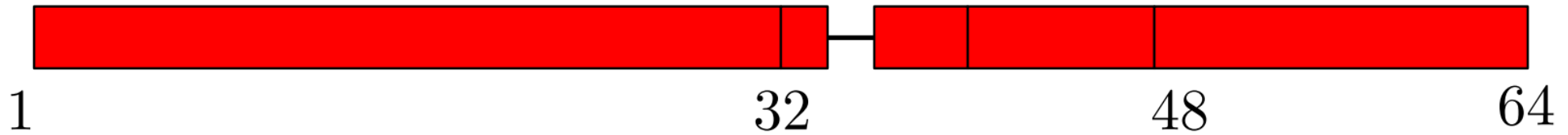
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Is $x > 36$? Answer: No

Is $x > 34$?

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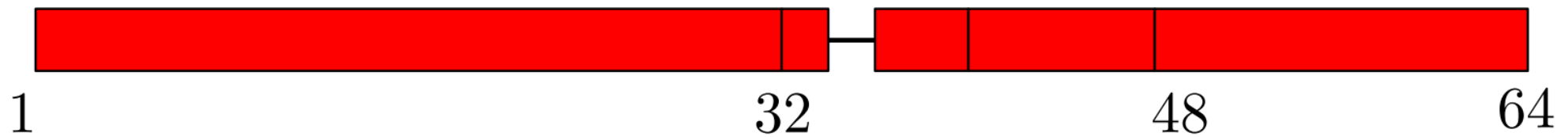
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Is $x > 34$? Answer: Yes

Binary Search Example



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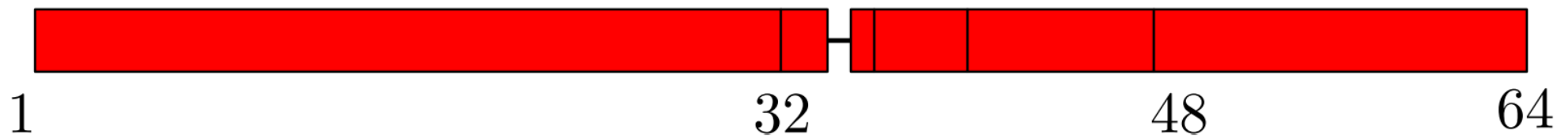
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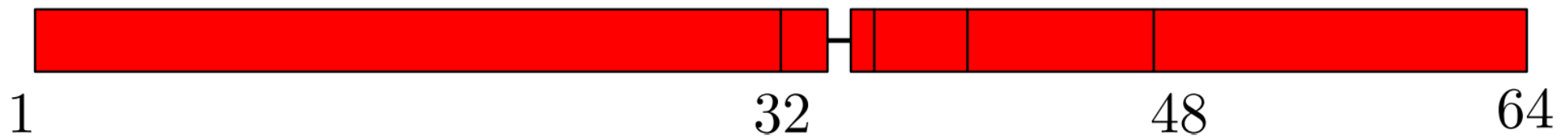
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Binary Search Example



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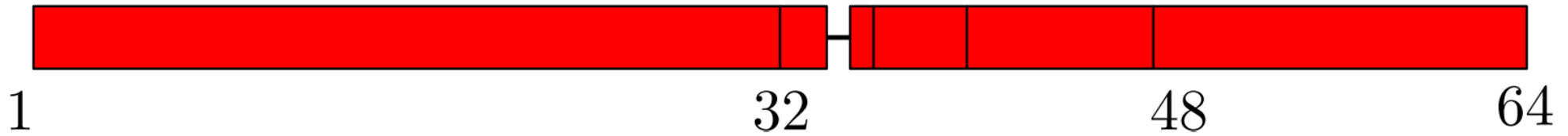
Is $x > 36$? Answer: No

Is $x > 34$? Answer: Yes

Is $x > 35$? Answer: No

Is $x = 35$?

Binary Search Example



Is $x > 32$?	Answer: Yes
Is $x > 48$?	Answer: No
Is $x > 40$?	Answer: No
Is $x > 36$?	Answer: No
Is $x > 34$?	Answer: Yes
Is $x > 35$?	Answer: No
Is $x = 35$?	Answer: BINGO!

Binary Search Example

- **Method**: Each guess **reduces** the problem to one in which the range is only **half** as big.



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Note: When n is a power of 2, $T(n)$, the number of questions in a binary search on $[1, n]$, satisfies

$$T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n/2) + 1 & \text{if } n \geq 2 \end{cases}$$



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This can also be proved **inductively**, similar to the tower of Hanoi recurrence.



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Base case (1 item): $T(1) = 1$ to ask: “**Is the number k ?**”



Binary Search Example

$$(*) \quad T(n) = \begin{cases} C_1 & \text{if } n = 1 \\ T(\lceil n/2 \rceil) + C_2 & \text{if } n \geq 2 \end{cases}$$

For simplicity, we will (usually) assume that n is a power of 2 (or sometimes 3 or 4) and also often that constants such as C_1, C_2 are 1. This will let us replace a recurrence such as $(*)$ by one such as $(**)$.



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In practice, the solution of (*) will be very close to that of (**) (this can be proved mathematically). Hence, we can restrict attention to (**).



Growth Rates of Solutions to Recurrences

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Iterating Recurrences: Example 1

■

$$(*) \quad T(n) = \begin{cases} T(1) & \text{if } n = 1 \\ 2T(n/2) + n & \text{if } n \geq 2 \end{cases}$$



Iterating Recurrences: Example 1

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This corresponds to solving a problem of size n , by

- (i) solving 2 subproblems of size $n/2$ and
- (ii) doing n units of additional work

or using $T(1)$ work for “bottom” case of $n = 1$



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We now see how to solve $(*)$ by algebraically iterating the recurrence.



Iterating Recurrences: Example 1

- Algebraically iterating the recurrence

$$T(n) = 2T\left(\frac{n}{2}\right) + n$$



Iterating Recurrences: Example 1

- Algebraically iterating the recurrence

Assume that n is a power of 2

$$T(n) = 2T\left(\frac{n}{2}\right) + n = 2\left(2T\left(\frac{n}{4}\right) + \frac{n}{2}\right) + n$$



Iterating Recurrences: Example 1

- Algebraically iterating the recurrence

Assume that n is a power of 2

$$\begin{aligned} T(n) &= 2T\left(\frac{n}{2}\right) + n &= 2\left(2T\left(\frac{n}{4}\right) + \frac{n}{2}\right) + n \\ &= 4T\left(\frac{n}{4}\right) + 2n &= 4\left(2T\left(\frac{n}{8}\right) + \frac{n}{4}\right) + 2n \end{aligned}$$



Iterating Recurrences: Example 1

- Algebraically iterating the recurrence

Assume that n is a power of 2

$$\begin{aligned}T(n) &= 2T\left(\frac{n}{2}\right) + n &= 2\left(2T\left(\frac{n}{4}\right) + \frac{n}{2}\right) + n \\&= 4T\left(\frac{n}{4}\right) + 2n &= 4\left(2T\left(\frac{n}{8}\right) + \frac{n}{4}\right) + 2n \\&= 8T\left(\frac{n}{8}\right) + 3n\end{aligned}$$



Iterating Recurrences: Example 1

- Algebraically iterating the recurrence

Assume that n is a power of 2

$$\begin{aligned}T(n) &= 2T\left(\frac{n}{2}\right) + n &&= 2\left(2T\left(\frac{n}{4}\right) + \frac{n}{2}\right) + n \\&= 4T\left(\frac{n}{4}\right) + 2n &&= 4\left(2T\left(\frac{n}{8}\right) + \frac{n}{4}\right) + 2n \\&= 8T\left(\frac{n}{8}\right) + 3n \\&\quad \vdots \quad \quad \quad \vdots \\&= 2^i T\left(\frac{n}{2^i}\right) + in\end{aligned}$$



Iterating Recurrences: Example 1

- Algebraically iterating the recurrence

Assume that n is a power of 2

$$T(n) = 2T\left(\frac{n}{2}\right) + n = 2\left(2T\left(\frac{n}{4}\right) + \frac{n}{2}\right) + n$$

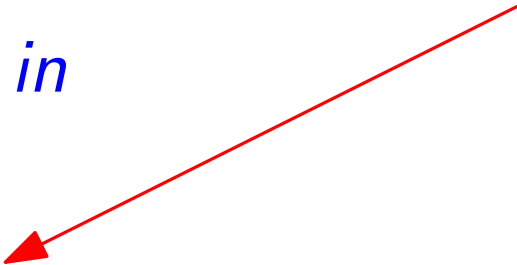
$$= 4T\left(\frac{n}{4}\right) + 2n = 4\left(2T\left(\frac{n}{8}\right) + \frac{n}{4}\right) + 2n$$

$$= 8T\left(\frac{n}{8}\right) + 3n$$

$$\vdots \quad \vdots$$
$$= 2^i T\left(\frac{n}{2^i}\right) + in$$

$$\vdots \quad \vdots$$
$$= 2^{\log_2 n} T\left(\frac{n}{2^{\log_2 n}}\right) + (\log_2 n)n$$

End when $i = \log_2 n$



Iterating Recurrences: Example 1

- Algebraically iterating the recurrence

Assume that n is a power of 2

$$T(n) = 2T\left(\frac{n}{2}\right) + n = 2\left(2T\left(\frac{n}{4}\right) + \frac{n}{2}\right) + n$$

$$= 4T\left(\frac{n}{4}\right) + 2n = 4\left(2T\left(\frac{n}{8}\right) + \frac{n}{4}\right) + 2n$$

$$= 8T\left(\frac{n}{8}\right) + 3n$$

$$\vdots \quad \vdots$$
$$= 2^i T\left(\frac{n}{2^i}\right) + in$$

End when $i = \log_2 n$

$$\vdots \quad \vdots$$
$$= 2^{\log_2 n} T\left(\frac{n}{2^{\log_2 n}}\right) + (\log_2 n)n$$

$$= nT(1) + n\log_2 n$$

Iterating Recurrences: Example 1

- We just iterated the recurrence to derive that the solution to

$$(*) \quad T(n) = \begin{cases} T(1) & \text{if } n = 1 \\ 2T(n/2) + n & \text{if } n \geq 2 \end{cases}$$

is $nT(1) + n \log_2 n$.



Iterating Recurrences: Example 1

- We just iterated the recurrence to derive that the solution to

$$(*) \quad T(n) = \begin{cases} T(1) & \text{if } n = 1 \\ 2T(n/2) + n & \text{if } n \geq 2 \end{cases}$$

is $nT(1) + n \log_2 n$.

Note: Technically, we still need to use **induction** to prove that our solution is correct. Practically, we **never** explicitly perform this step, since it is obvious how the induction would work.



Iterating Recurrences: Example 2

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$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n/2) + 1 & \text{if } n \geq 2 \end{cases}$$

Iterating Recurrences: Example 2



$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n/2) + 1 & \text{if } n \geq 2 \end{cases}$$

$$T(n) = T\left(\frac{n}{2}\right) + 1$$



Iterating Recurrences: Example 2



$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n/2) + 1 & \text{if } n \geq 2 \end{cases}$$

$$T(n) = T\left(\frac{n}{2}\right) + 1 = \left(T\left(\frac{n}{2^2}\right) + 1\right) + 1$$



Iterating Recurrences: Example 2



$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n/2) + 1 & \text{if } n \geq 2 \end{cases}$$

$$\begin{aligned} T(n) &= T\left(\frac{n}{2}\right) + 1 &= \left(T\left(\frac{n}{2^2}\right) + 1\right) + 1 \\ &= T\left(\frac{n}{2^2}\right) + 2 \end{aligned}$$



Iterating Recurrences: Example 2



$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n/2) + 1 & \text{if } n \geq 2 \end{cases}$$

$$\begin{aligned} T(n) &= T\left(\frac{n}{2}\right) + 1 &= \left(T\left(\frac{n}{2^2}\right) + 1\right) + 1 \\ &= T\left(\frac{n}{2^2}\right) + 2 &= \left(T\left(\frac{n}{2^3}\right) + 1\right) + 2 \end{aligned}$$



Iterating Recurrences: Example 2



$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n/2) + 1 & \text{if } n \geq 2 \end{cases}$$

$$\begin{aligned} T(n) &= T\left(\frac{n}{2}\right) + 1 &= \left(T\left(\frac{n}{2^2}\right) + 1\right) + 1 \\ &= T\left(\frac{n}{2^2}\right) + 2 &= \left(T\left(\frac{n}{2^3}\right) + 1\right) + 2 \\ &= T\left(\frac{n}{2^3}\right) + 3 \end{aligned}$$



Iterating Recurrences: Example 2



$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n/2) + 1 & \text{if } n \geq 2 \end{cases}$$

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$$= T\left(\frac{n}{2^2}\right) + 2 = \left(T\left(\frac{n}{2^3}\right) + 1\right) + 2$$

$$= T\left(\frac{n}{2^3}\right) + 3$$

$$\vdots \quad \vdots$$

$$= T\left(\frac{n}{2^i}\right) + i$$



Iterating Recurrences: Example 2

$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n/2) + 1 & \text{if } n \geq 2 \end{cases}$$

$$T(n) = T\left(\frac{n}{2}\right) + 1 = \left(T\left(\frac{n}{2^2}\right) + 1\right) + 1$$

$$= T\left(\frac{n}{2^2}\right) + 2 = \left(T\left(\frac{n}{2^3}\right) + 1\right) + 2$$

$$= T\left(\frac{n}{2^3}\right) + 3$$

$$\begin{array}{c} \vdots \\ \vdots \\ = T\left(\frac{n}{2^i}\right) + i \end{array}$$

$$\begin{array}{c} \vdots \\ \vdots \\ = T\left(\frac{n}{2^{\log_2 n}}\right) + \log_2 n \end{array}$$



Iterating Recurrences: Example 2



$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n/2) + 1 & \text{if } n \geq 2 \end{cases}$$

$$T(n) = T\left(\frac{n}{2}\right) + 1 = \left(T\left(\frac{n}{2^2}\right) + 1\right) + 1$$

$$= T\left(\frac{n}{2^2}\right) + 2 = \left(T\left(\frac{n}{2^3}\right) + 1\right) + 2$$

$$= T\left(\frac{n}{2^3}\right) + 3$$

$$\begin{array}{c} \vdots \\ \vdots \\ = T\left(\frac{n}{2^i}\right) + i \end{array}$$

$$\begin{array}{c} \vdots \\ \vdots \\ = T\left(\frac{n}{2^{\log_2 n}}\right) + \log_2 n = 1 + \log_2 n \end{array}$$



Iterating Recurrences: Example 3

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$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n/2) + n & \text{if } n \geq 2 \end{cases}$$



Iterating Recurrences: Example 3



$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n/2) + n & \text{if } n \geq 2 \end{cases}$$

$$\begin{aligned} T(n) &= T\left(\frac{n}{2}\right) + n \\ &= T\left(\frac{n}{2^2}\right) + \frac{n}{2} + n \end{aligned}$$



Iterating Recurrences: Example 3



$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n/2) + n & \text{if } n \geq 2 \end{cases}$$

$$T(n) = T\left(\frac{n}{2}\right) + n$$

$$= T\left(\frac{n}{2^2}\right) + \frac{n}{2} + n$$

$$= T\left(\frac{n}{2^3}\right) + \frac{n}{2^2} + \frac{n}{2} + n$$



Iterating Recurrences: Example 3

■

$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n/2) + n & \text{if } n \geq 2 \end{cases}$$

$$\begin{aligned} T(n) &= T\left(\frac{n}{2}\right) + n \\ &= T\left(\frac{n}{2^2}\right) + \frac{n}{2} + n \\ &= T\left(\frac{n}{2^3}\right) + \frac{n}{2^2} + \frac{n}{2} + n \\ &\quad \vdots \\ &= T\left(\frac{n}{2^i}\right) + \frac{n}{2^{i-1}} + \cdots + \frac{n}{2^2} + \frac{n}{2} + n \end{aligned}$$

Iterating Recurrences: Example 3

$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n/2) + n & \text{if } n \geq 2 \end{cases}$$

$$T(n) = T\left(\frac{n}{2}\right) + n$$

$$= T\left(\frac{n}{2^2}\right) + \frac{n}{2} + n$$

$$= T\left(\frac{n}{2^3}\right) + \frac{n}{2^2} + \frac{n}{2} + n$$

$$\begin{array}{c} \vdots \\ \vdots \\ = T\left(\frac{n}{2^i}\right) + \frac{n}{2^{i-1}} + \cdots + \frac{n}{2^2} + \frac{n}{2} + n \end{array}$$

$$\begin{array}{c} \vdots \\ \vdots \\ = T\left(\frac{n}{2^{\log_2 n}}\right) + \frac{n}{2^{\log_2 n - 1}} + \cdots + \frac{n}{2^2} + \frac{n}{2} + n \end{array}$$

Iterating Recurrences: Example 3

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$$= T\left(\frac{n}{2^3}\right) + \frac{n}{2^2} + \frac{n}{2} + n$$

$$\begin{array}{c} \vdots \\ \vdots \\ = T\left(\frac{n}{2^i}\right) + \frac{n}{2^{i-1}} + \cdots + \frac{n}{2^2} + \frac{n}{2} + n \end{array}$$

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$$= 1 + 2 + 2^2 + \cdots + \frac{n}{2^2} + \frac{n}{2} + n$$



Iterating Recurrences: Example 3

$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n/2) + n & \text{if } n \geq 2 \end{cases}$$

$$T(n) = T\left(\frac{n}{2}\right) + n$$

$$= T\left(\frac{n}{2^2}\right) + \frac{n}{2} + n$$

$$= T\left(\frac{n}{2^3}\right) + \frac{n}{2^2} + \frac{n}{2} + n$$

$$\vdots \quad \vdots$$

$$= T\left(\frac{n}{2^i}\right) + \frac{n}{2^{i-1}} + \cdots + \frac{n}{2^2} + \frac{n}{2} + n$$

$$\vdots \quad \vdots$$

$$= T\left(\frac{n}{2^{\log_2 n}}\right) + \frac{n}{2^{\log_2 n - 1}} + \cdots + \frac{n}{2^2} + \frac{n}{2} + n$$

$$= 1 + 2 + 2^2 + \cdots + \frac{n}{2^2} + \frac{n}{2} + n = \Theta(n)$$

Iterating Recurrences: Example 4

■

$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n < 3 \\ 3T(n/3) + n & \text{if } n \geq 3 \end{cases}$$



Iterating Recurrences: Example 4

■

$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n < 3 \\ 3T(n/3) + n & \text{if } n \geq 3 \end{cases}$$

$$T(n) = 3T\left(\frac{n}{3}\right) + n$$



Iterating Recurrences: Example 4

■

$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n < 3 \\ 3T(n/3) + n & \text{if } n \geq 3 \end{cases}$$

$$T(n) = 3T\left(\frac{n}{3}\right) + n = 3\left(3T\left(\frac{n}{3^2}\right) + \frac{n}{3}\right) + n$$



Iterating Recurrences: Example 4

■

$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n < 3 \\ 3T(n/3) + n & \text{if } n \geq 3 \end{cases}$$

$$\begin{aligned} T(n) &= 3T\left(\frac{n}{3}\right) + n &= 3\left(3T\left(\frac{n}{3^2}\right) + \frac{n}{3}\right) + n \\ &= 3^2 T\left(\frac{n}{3^2}\right) + 2n \end{aligned}$$



Iterating Recurrences: Example 4

■

$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n < 3 \\ 3T(n/3) + n & \text{if } n \geq 3 \end{cases}$$

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Iterating Recurrences: Example 4

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$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n < 3 \\ 3T(n/3) + n & \text{if } n \geq 3 \end{cases}$$

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Iterating Recurrences: Example 4

■

$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n < 3 \\ 3T(n/3) + n & \text{if } n \geq 3 \end{cases}$$

$$\begin{aligned} T(n) &= 3T\left(\frac{n}{3}\right) + n &= 3\left(3T\left(\frac{n}{3^2}\right) + \frac{n}{3}\right) + n \\ &= 3^2 T\left(\frac{n}{3^2}\right) + 2n &= 3^2\left(3T\left(\frac{n}{3^3}\right) + \frac{n}{3^2}\right) + 2n \\ &= 3^3 T\left(\frac{n}{3^3}\right) + 3n \\ &\quad \vdots \quad \vdots \\ &= 3^i T\left(\frac{n}{3^i}\right) + in \\ &\quad \vdots \quad \vdots \\ &= 3^{\log_3 n} T\left(\frac{n}{3^{\log_3 n}}\right) + n \log_3 n \end{aligned}$$



Iterating Recurrences: Example 4

$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n < 3 \\ 3T(n/3) + n & \text{if } n \geq 3 \end{cases}$$

$$\begin{aligned} T(n) &= 3T\left(\frac{n}{3}\right) + n &= 3\left(3T\left(\frac{n}{3^2}\right) + \frac{n}{3}\right) + n \\ &= 3^2 T\left(\frac{n}{3^2}\right) + 2n &= 3^2\left(3T\left(\frac{n}{3^3}\right) + \frac{n}{3^2}\right) + 2n \\ &= 3^3 T\left(\frac{n}{3^3}\right) + 3n \\ &\quad \vdots \quad \vdots \\ &= 3^i T\left(\frac{n}{3^i}\right) + in \\ &\quad \vdots \quad \vdots \\ &= 3^{\log_3 n} T\left(\frac{n}{3^{\log_3 n}}\right) + n \log_3 n = n + n \log_3 n \end{aligned}$$



Iterating Recurrences: Example 5

■

$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ 4T(n/2) + n & \text{if } n \geq 2 \end{cases}$$



Iterating Recurrences: Example 5

■

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$$T(n) = 4T\left(\frac{n}{2}\right) + n$$



Iterating Recurrences: Example 5

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$$T(n) = 4T\left(\frac{n}{2}\right) + n = 4\left(4T\left(\frac{n}{2^2}\right) + \frac{n}{2}\right) + n$$



Iterating Recurrences: Example 5

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$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ 4T(n/2) + n & \text{if } n \geq 2 \end{cases}$$

$$\begin{aligned} T(n) &= 4T\left(\frac{n}{2}\right) + n &= 4\left(4T\left(\frac{n}{2^2}\right) + \frac{n}{2}\right) + n \\ &= 4^2 T\left(\frac{n}{2^2}\right) + \frac{4}{2}n + n \end{aligned}$$



Iterating Recurrences: Example 5

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$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ 4T(n/2) + n & \text{if } n \geq 2 \end{cases}$$

$$\begin{aligned} T(n) &= 4T\left(\frac{n}{2}\right) + n &= 4\left(4T\left(\frac{n}{2^2}\right) + \frac{n}{2}\right) + n \\ &= 4^2 T\left(\frac{n}{2^2}\right) + \frac{4}{2}n + n &= 4^2\left(4T\left(\frac{n}{2^3}\right) + \frac{n}{2^2}\right) + \frac{4}{2}n + n \end{aligned}$$



Iterating Recurrences: Example 5

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$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ 4T(n/2) + n & \text{if } n \geq 2 \end{cases}$$

$$\begin{aligned} T(n) &= 4T\left(\frac{n}{2}\right) + n &&= 4\left(4T\left(\frac{n}{2^2}\right) + \frac{n}{2}\right) + n \\ &= 4^2 T\left(\frac{n}{2^2}\right) + \frac{4}{2}n + n &&= 4^2\left(4T\left(\frac{n}{2^3}\right) + \frac{n}{2^2}\right) + \frac{4}{2}n + n \\ &= 4^3 T\left(\frac{n}{2^3}\right) + \frac{4^2}{2^2}n + \frac{4}{2}n + n \end{aligned}$$



Iterating Recurrences: Example 5

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$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ 4T(n/2) + n & \text{if } n \geq 2 \end{cases}$$

$$\begin{aligned} T(n) &= 4T\left(\frac{n}{2}\right) + n &&= 4\left(4T\left(\frac{n}{2^2}\right) + \frac{n}{2}\right) + n \\ &= 4^2 T\left(\frac{n}{2^2}\right) + \frac{4}{2}n + n &&= 4^2\left(4T\left(\frac{n}{2^3}\right) + \frac{n}{2^2}\right) + \frac{4}{2}n + n \\ &= 4^3 T\left(\frac{n}{2^3}\right) + \frac{4^2}{2^2}n + \frac{4}{2}n + n \\ &\quad \vdots \quad \quad \quad \vdots \\ &= 4^i T\left(\frac{n}{2^i}\right) + \frac{4^{i-1}}{2^{i-1}}n + \cdots + \frac{4^2}{2^2}n + n \end{aligned}$$

Iterating Recurrences: Example 5

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$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ 4T(n/2) + n & \text{if } n \geq 2 \end{cases}$$

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Iterating Recurrences: Example 5

$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ 4T(n/2) + n & \text{if } n \geq 2 \end{cases}$$

$$\begin{aligned} T(n) &= 4T\left(\frac{n}{2}\right) + n &&= 4\left(4T\left(\frac{n}{2^2}\right) + \frac{n}{2}\right) + n \\ &= 4^2 T\left(\frac{n}{2^2}\right) + \frac{4}{2}n + n &&= 4^2\left(4T\left(\frac{n}{2^3}\right) + \frac{n}{2^2}\right) + \frac{4}{2}n + n \\ &= 4^3 T\left(\frac{n}{2^3}\right) + \frac{4^2}{2^2}n + \frac{4}{2}n + n \\ &\quad \vdots \quad \vdots \\ &= 4^i T\left(\frac{n}{2^i}\right) + \frac{4^{i-1}}{2^{i-1}}n + \cdots + \frac{4^2}{2^2}n + n \\ &\quad \vdots \quad \vdots \\ &= 4^{\log_2 n} T\left(\frac{n}{2^{\log_2 n}}\right) + \frac{4^{\log_2 n - 1}}{2^{\log_2 n - 1}}n + \cdots + \frac{4}{2}n + n \\ &= 2n^2 - n \end{aligned}$$



Three Different Behaviors

- Compare the iteration for the recurrences

$$T(n) = 2T(n/2) + n$$

$$T(n) = T(n/2) + n$$

$$T(n) = 4T(n/2) + n$$



Three Different Behaviors

- Compare the iteration for the recurrences

$$T(n) = 2T(n/2) + n$$

$$T(n) = T(n/2) + n$$

$$T(n) = 4T(n/2) + n$$

- ◇ all three recurrences iterate $\log_2 n$ times
- ◇ in each case, size of subproblem in next iteration is **half** the size in the preceding iteration level



Three Different Behaviors

- **Theorem** Suppose that we have a recurrence of the form

$$T(n) = aT(n/2) + n,$$

where a is a positive integer and $T(1)$ is nonnegative. Then we have the following **big Θ** bounds on the solution:

1. If $a < 2$, then $T(n) = \Theta(n)$.
2. If $a = 2$, then $T(n) = \Theta(n \log n)$.
3. If $a > 2$, then $T(n) = \Theta(n^{\log_2 a})$.



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Proof

We already proved Case 1 when $a = 1$ in Example 3.
(will not prove it for $1 < a < 2$)

We already proved Case 2 in Example 1.

We will now prove Case 3.



Iterating Recurrences

- $T(n) = aT(n/2) + n$, where $a > 2$. Assume that $n = 2^i$.



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Iterating as in Example 5 gives

$$T(n) = a^i T\left(\frac{n}{2^i}\right) + \left(\frac{a^{i-1}}{2^{i-1}} + \frac{a^{i-2}}{2^{i-2}} + \cdots + \frac{a}{2} + 1\right) n$$



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$$T(n) = a^{\log_2 n} T(1) + n \sum_{i=0}^{\log_2 n - 1} \left(\frac{a}{2}\right)^i$$

Work at
“bottom”

Iterated
Work



Total work

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Since $a > 2$, the geometric series is Θ of the largest term.

$$n \sum_{i=0}^{\log_2 n - 1} \left(\frac{a}{2}\right)^i = n \frac{1 - (a/2)^{\log_2 n}}{1 - a/2} = n\Theta((a/2)^{\log_2 n - 1})$$



Total work

- n times the largest term in the geometric series is

$$n \left(\frac{a}{2}\right)^{\log_2 n - 1} = \frac{2}{a} \cdot \frac{n \cdot a^{\log_2 n}}{2^{\log_2 n}} = \frac{2}{a} \cdot \frac{n \cdot a^{\log_2 n}}{n} = \frac{2}{a} \cdot a^{\log_2 n}$$



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Notice that

$$a^{\log_2 n} = (2^{\log_2 a})^{\log_2 n} = (2^{\log_2 n})^{\log_2 a} = n^{\log_2 a}$$

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$$a^{\log_2 n} T(1) + n \sum_{i=0}^{\log_2 n - 1} \left(\frac{a}{2}\right)^i$$



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$$\Theta \left(n^{\log_2 a} \right)$$

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Example 5 Recap

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$$(*) \quad T(n) = \begin{cases} 1 & \text{if } n = 1 \\ 4T(n/2) + n & \text{if } n \geq 2 \end{cases}$$



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$a = 4$, so the Theorem says that

$$T(n) = \Theta(n^{\log_2 a}) = \Theta(n^{\log_2 4}) = \Theta(n^2)$$



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$a = 4$, so the Theorem says that

$$T(n) = \Theta(n^{\log_2 a}) = \Theta(n^{\log_2 4}) = \Theta(n^2)$$

This matches with the exact answer of $2n^2 - n$.

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The Master Theorem

- **Theorem** Suppose that we have a recurrence of the form

$$T(n) = aT(n/b) + cn^d,$$

where a is a positive integer, $b \geq 1$, c, d are real numbers with c positive and d nonnegative, and $T(1)$ is nonnegative. Then we have the following **big Θ** bounds on the solution:

1. If $a < b^d$, then $T(n) = \Theta(n^d)$.
2. If $a = b^d$, then $T(n) = \Theta(n^d \log n)$.
3. If $a > b^d$, then $T(n) = \Theta(n^{\log_b a})$.



Next Lecture

- counting ...

